

Anti-derivative (Integration):

Let $f(x)$ and $F(x)$ be two functions of x such that $\frac{d}{dx}(F(x)) = f(x)$, then $F(x)$ is called an anti-derivative of $f(x)$ or an integral of $f(x)$ with respect to x and written as, $\int f(x)dx = F(x) + C$, Where C is an arbitrary constant called constant of integration.

Integrals of Some Elementary Functions:

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C; n \neq -1$
2. $\int (ax + b)^n dx = \frac{(ax+b)^{n+1}}{a.(n+1)} + C; n \neq -1$
3. $\int 1. dx = x + C$
4. $\int k. dx = k \int 1. dx = kx + C$
5. $\int \frac{1}{x} dx = \ln|x| + C$
6. $\int \frac{1}{ax+b} dx = \frac{\ln|ax+b|}{a} + C$
7. $\int e^x dx = e^x + C$
8. $\int e^{ax+b} dx = \frac{e^{ax+b}}{a} + C$
9. $\int a^x dx = \frac{a^x}{\ln|a|} + C$
10. $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
11. $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$
12. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$

Integrals of Some Trigonometric Functions

1. $\int \sin x dx = -\cos x + C$
2. $\int \cos x dx = \sin x + C$
3. $\int \sec^2 x dx = \tan x + C$
4. $\int \operatorname{cosec}^2 x dx = -\cot x + C$
5. $\int \sec x . \tan x dx = \sec x + C$
6. $\int \operatorname{cosec} x . \cot x dx = -\operatorname{cosec} x + C$
7. $\int \tan x dx = \ln | \tan x | + C = -\ln | \cos x | + C$
8. $\int \cot x dx = \ln | \sin x | + C$
9. $\int \operatorname{cosec} x dx = \ln | \operatorname{cosec} x - \cot x | + C$
10. $\int \sec x dx = \ln | \sec x + \tan x | + C$

Rules of integration:

1. Coefficient rule:

$$\int k. f(x) dx = k \int f(x) dx$$

2. Sum/ difference rule:

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

3. Integration by parts:

$$\int u. v dx = u \int v. dx - \int \left[\frac{d}{dx}(u). \int v dx \right] dx$$

Standard Integrals (I):

$$1. \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$\begin{aligned}
 2. \int \frac{1}{a^2-x^2} dx &= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C \\
 3. \int \frac{1}{x^2-a^2} dx &= \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \\
 4. \int \frac{1}{\sqrt{a^2-x^2}} dx &= \sin^{-1} \frac{x}{a} + C \\
 5. \int \frac{1}{\sqrt{a^2+x^2}} dx &= \ln |x + \sqrt{a^2+x^2}| + C = \sinh^{-1} \frac{x}{a} + C \\
 6. \int \frac{1}{\sqrt{x^2-a^2}} dx &= \ln |x + \sqrt{x^2-a^2}| + C = \cosh^{-1} \frac{x}{a} + C
 \end{aligned}$$

Q. Prove that:

$$\begin{aligned}
 \text{i) } \int \frac{1}{\sqrt{a^2-x^2}} dx &= \sin^{-1} \frac{x}{a} + C \\
 \text{ii) } \int \frac{1}{a^2+x^2} dx &= \frac{1}{a} \tan^{-1} \frac{x}{a} + C \\
 \text{iii) } \int \frac{1}{a^2-x^2} dx &= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C \\
 &= \sinh^{-1} \frac{x}{a} + C \\
 \text{iv) } \int \frac{1}{\sqrt{a^2+x^2}} dx &= \ln |x + \sqrt{a^2+x^2}| + C
 \end{aligned}$$

Integrals Reducible to Standard Integral I

Type: 1

Integrals of the form $\int \frac{1}{ax^2+bx+c} dx$ or $\int \frac{1}{\sqrt{ax^2+bx+c}} dx$

Rule: Make the coeff. of x^2 unity and complete the square in terms containing x^2 and x and then use standard integral I

Q. Evaluate:

$$\begin{aligned}
 1. \int \frac{dx}{x^2-16} \quad & 2. \int \frac{dx}{4x^2-9} \quad & 3. \int \frac{dx}{a^2-b^2x^2} \quad & 4. \int \frac{dx}{2x^2+x-1} \quad & 5. \int \frac{dx}{1+x-x^2} \quad & 6. \int \frac{dx}{3-2x-x^2} \\
 7. \int \frac{dx}{\sqrt{9x^2-1}} \quad & 8. \int \frac{dx}{\sqrt{2ax+x^2}} \quad & 9. \int \frac{dx}{\sqrt{x^2-6x+13}} \quad & 10. \int \frac{dx}{\sqrt{15-4x-4x^2}}
 \end{aligned}$$

Answers:

$$\begin{aligned}
 1. \frac{1}{8} \ln \left| \frac{x-4}{x+4} \right| + C \quad & 2. \frac{1}{12} \ln \left| \frac{2x-3}{2x+3} \right| + C \\
 3. \frac{1}{2ab} \ln \left| \frac{a+bx}{a-bx} \right| + C \quad & 4. \frac{1}{3} \ln \left| \frac{2x-1}{2(x+1)} \right| + C \\
 5. \frac{1}{\sqrt{5}} \ln \left| \frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x} \right| + C \quad & 6. \frac{1}{4} \ln \left| \frac{3+x}{1-x} \right| + C \\
 7. \frac{1}{3} \ln |3x + \sqrt{9x^2-1}| + C \\
 8. \ln |x + a + \sqrt{2ax+x^2}| + C \\
 9. \ln |x - 3 + \sqrt{x^2-6x+13}| + C \\
 10. \frac{1}{2} \sin^{-1} \left(\frac{2x+1}{4} \right) + C
 \end{aligned}$$

Type: 2

Q. Evaluate:

$$\begin{aligned}
 1. \int \frac{dx}{e^x+e^{-x}} \quad & 2. \int \frac{dx}{\sqrt{1+e^{-2x}}} \quad & 3. \int \frac{e^x dx}{e^{2x}-4} \\
 4. \int \frac{2^x dx}{\sqrt{1-4^x}} \quad & 5. \int \frac{x dx}{\sqrt{x^4+1}} \quad & 6. \int \frac{x dx}{\sqrt{3x^2+1}} \\
 7. \int \frac{\cos x dx}{\sqrt{\sin^2 x - 2 \sin x - 3}} \quad & 8. \int \frac{\sec^2 x dx}{\sqrt{\tan^2 x + 4}}
 \end{aligned}$$

9. $\int \frac{x^2 dx}{1-x^6}$

10. $\int \frac{x dx}{\sqrt{x^4+2x^2+5}}$

Answers:

1. $\tan^{-1}(e^x) + C$ 2. $\ln|e^x + \sqrt{e^{2x} + 1}| + C$

3. $\frac{1}{4} \ln \left| \frac{e^x-2}{e^x+2} \right| + C$ 4. $\frac{1}{\ln 2} \sin^{-1}(2^x) + C$

5. $\frac{1}{2} \ln|x^2 + \sqrt{x^4 + 1}| + C$ 6. $\frac{1}{3} \ln|\sqrt{3x^2 + 4}| + C$ 7. $\ln|\sin x - 1 + \sqrt{\sin^2 x - 2\sin x - 3}| + C$

8. $\ln|\tan x + \sqrt{\tan^2 x + 4}| + C$ 9. $\frac{1}{6} \ln \left| \frac{1+x^3}{1-x^3} \right| + c$

10. $\frac{1}{2} \ln|x^2 + 1 + \sqrt{x^4 + 2x^2 + 5}| + C$

Type-3

Integrals of the form $\int \frac{(px+q)}{ax^2+bx+c} dx$ or $\int \frac{(px+q)}{\sqrt{ax^2+bx+c}} dx$

Rule: Change $px + q$ into the derivative of $ax^2 + bx + c$ and express the integral into the sum of two integrals and change second integral into standard integral I as type 1.

Q. Evaluate:

1. $\int \frac{2x-3}{x^2+3x-18} dx$

2. $\int \frac{2x+2}{3+2x-x^2} dx$

3. $\int \frac{x-1}{(x-2)(x-3)} dx$

4. $\int \frac{6x+1}{x^2+9} dx$

5. $\int \frac{2x-11}{x^2+x-2} dx$

6. $\int \frac{x dx}{\sqrt{x^2+4x+5}}$

7. $\int \frac{x+2}{\sqrt{x^2+2x-1}} dx$

8. $\int \frac{2x-8}{\sqrt{1-x-x^2}} dx$

Answers:

1. $\ln|x^2 + 3x - 18| - \frac{2}{3} \ln \left| \frac{x-3}{x+6} \right| + C$

2. $\ln \left| \frac{x+1}{3-x} \right| - \ln|3 + 2x - x^2| + C$

3. $\frac{1}{2} \ln|x^2 - 5x + 6| + \frac{3}{2} \ln \left| \frac{x-3}{x-2} \right| + C$

4. $3 \ln|x^2 + 9| + \frac{1}{3} \tan^{-1} \frac{x}{3} + C$

5. $5 \ln|x + 2| - 3 \ln|x - 1| + C$

6. $\sqrt{x^2 + 4x + 5} - 2 \ln|x + 2 + \sqrt{x^2 + 4x + 5}| + C$ 7. $\sqrt{x^2 + 2x - 1} + \ln|x + 1 + \sqrt{x^2 + 2x - 1}| + C$

8. $-2\sqrt{1-x-x^2} - 9 \sin^{-1} \frac{2x+1}{\sqrt{5}} + C$

Type-4:

Integrals of the form : $\int \frac{dx}{(ax+b)\sqrt{cx+d}}$

Rule: Put $y^2 = cx + d$ then change the whole integral in terms of y

Q. Evaluate:

1. $\int \frac{dx}{(4x+3)\sqrt{x+3}}$

2. $\int \frac{dx}{(2x-1)\sqrt{4x-3}}$

3. $\int \frac{dx}{(2x+3)\sqrt{4x+5}}$

Answers:

1. $\frac{1}{6} \ln \left| \frac{2\sqrt{x+3}-3}{2\sqrt{x+3}+3} \right| + C$

2. $4 \tan^{-1}(\sqrt{4x-3}) + C$

3. $\tan^{-1}(\sqrt{4x+5}) + C$

Type-5:

Integrals of the form: $\int \frac{dx}{\sqrt{(ax+b)(cx+d)}}$

Rule: Put $y^2 = (ax + b)$ or $(cx + d)$ then change the whole integral in terms of y

$$1. \int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}} \quad (\beta > \alpha) \quad 2. \int \frac{dx}{\sqrt{(x+\alpha)(x+\beta)}}$$

$$3. \int \frac{dx}{\sqrt{(x+5)(x-1)}}$$

Answers:

$$1. 2\ln \left| \sqrt{(x-\alpha)} + \sqrt{(x-\beta)} \right| + C$$

$$2. 2\ln \left| \sqrt{(x+\alpha)} + \sqrt{(x+\beta)} \right| + C$$

$$3. 2\ln \left| \sqrt{(x+5)} + \sqrt{(x-1)} \right| + C$$

Type- 6:

Integrals of the form $\int \sqrt{\frac{ax+b}{cx+d}} dx$

Rule: Rationalize the numerator (i.e multiply both N^r and D^r by $ax + b$) then it will be converted to type 3.

Q. Evaluate:

$$1. \int \sqrt{\frac{a+x}{a-x}} dx \quad 2. \int \sqrt{\frac{1+x}{1-x}} dx \quad 3. \int \sqrt{\frac{1+x}{x}} dx$$

Answers:

$$1. a \sin^{-1} \frac{x}{a} - \sqrt{a^2 - x^2} + C$$

$$2. \sin^{-1} x - \sqrt{1 - x^2} + C$$

$$3. \sqrt{x + x^2} + \frac{1}{2} \ln \left| x + \frac{1}{2} + \sqrt{x + x^2} \right| + C$$

Some extra questions:

1. Define integration. Why do we add C after integration?
2. Derivative and integration are the reverse process. Justify the statement with suitable example.
3. How do you reduce the integrals of the form $\int \frac{(px+q)}{ax^2+bx+c} dx$ into standard integral I ?

Standard Integral II:

$$1. \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln | x + \sqrt{x^2 + a^2} | + C$$

$$2. \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln | x + \sqrt{x^2 - a^2} | + C$$

$$3. \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

Integrals Reducible to Standard Integral II

1. Integrals of the form $\int \sqrt{ax^2 + bx + c} dx$

Rule: Make the coeff. of x^2 unity and change it into standard integral II

2. Integrals of the form $\int (px + q) \sqrt{ax^2 + bx + c} dx$

Rule: Change $px + q$ into the derivative of $ax^2 + bx + c$ and express the integral into the sum of two integrals and change second integral into standard integral II

Note: $\int f'(x) \cdot \sqrt{f(x)} dx = \frac{2}{3} [f(x)]^{\frac{3}{2}} + C$

Questions:

1. Evaluate:

a) $\int \sqrt{x^2 - 5} dx$

Ans: $\frac{x}{2} \sqrt{x^2 - 5} - \frac{5}{2} \ln | x + \sqrt{x^2 - 5} | + C$

b) $\int \sqrt{25 - 9x^2} dx$

Ans: $\frac{1}{6} \left(3x \sqrt{25 - 9x^2} + 25 \sin^{-1} \frac{3x}{5} \right) + C$

c) $\int \sqrt{2ax - x^2} dx$

Ans: $\frac{1}{2} (x - a) \sqrt{2ax - x^2} + \frac{1}{2} a^2 \sin^{-1} \left(\frac{x-a}{a} \right) + c$

d) $\int \sqrt{9x^2 + 24x + 25} dx$

Ans: $\frac{1}{6} (3x + 4) \sqrt{9x^2 + 24x + 25} + \frac{3}{2} \ln |3x + 4 + \sqrt{9x^2 + 24x + 25}| + c$

e) $\int \sqrt{(x-a)(b-x)} dx$

Ans: $\frac{1}{4} (2x - a - b) \sqrt{(x-a)(b-x)} + \frac{1}{8} (a-b)^2 \sin^{-1} \left(\frac{2x-a-b}{a-b} \right) + c$

2. Evaluate:

a) $\int (2x - 5) \sqrt{x^2 - 4x + 5} dx$

Ans: $\frac{2}{3} (x^2 - 4x + 5)^{\frac{3}{2}} - \frac{(x-2)}{2} \sqrt{x^2 - 4x + 5} - \frac{1}{2} \ln |x - 2 + \sqrt{x^2 - 4x + 5}| + c$

b) $\int (2 - x) \sqrt{16 - 6x - x^2} dx$

Ans: $\frac{1}{3} (16 - 6x - x^2)^{\frac{3}{2}} + 5 \frac{(x+3)}{2} \sqrt{16 - 6x - x^2} + \frac{125}{2} \sin^{-1} \left(\frac{x+3}{5} \right) + c$

3. Evaluate:

a) $\int \frac{dx}{x + \sqrt{x^2 - 1}}$

Ans: $\frac{x^2}{2} - \frac{x}{2} \sqrt{x^2 - 1} + \frac{1}{2} \ln |x + \sqrt{x^2 - 1}| + c$

b) $\int \frac{dx}{\sqrt{x^2 + 4} - x}$

Ans: $\frac{x^2}{8} + \frac{x}{8} \sqrt{x^2 + 4} + \frac{1}{2} \ln |x + \sqrt{x^2 + 4}| + c$

Standard Integral III

1. $\int \tan x dx = \ln |\tan x| + C = -\ln |\cos x| + C$

2. $\int \cot x dx = \ln |\sin x| + C$

3. $\int \operatorname{cosec} x dx = \ln \left| \tan \frac{x}{2} \right| + C$
 $= \ln |\operatorname{cosec} x - \cot x| + C$

4. $\int \sec x dx = \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C$
 $= \ln |\sec x + \tan x| + C$

Integrals Reducible to Standard Forms:

1. Integrals of the form $\int \frac{dx}{a \cos^2 x + b \sin^2 x}$; $\int \frac{dx}{a + b \cos^2 x}$; $\int \frac{dx}{a \cos^2 x + b \sin x \cos x + c \sin^2 x}$

Hints: Multiply N^r and D^r by $\sec^2 x$ and put $y = \tan x$

2. Integrals of the form $\int \frac{dx}{a \cos x + b \sin x}$; $\int \frac{dx}{a + b \cos x}$; $\int \frac{dx}{a + b \sin x}$; $\int \frac{dx}{a \cos x + b \sin x + c}$

Hints: Multiply N^r and D^r by $\frac{\sec^2 x}{2}$ and put

$y = \tan \frac{x}{2}$

1. $\int \sin hx dx = \cosh x + C$

$\int \cosh x dx = \sinh x + C$

3. $\int \sec^2 hx dx = \tanh x + C$

$\int \operatorname{cosec}^2 hx dx = -\coth x + C$

Integrals of Hyperbolic Functions:

2.

4.

$$5. \int \operatorname{sech} x \cdot \tanh x \, dx = -\operatorname{sech} x + C$$

$$\int \operatorname{cosec} x \cdot \cot x \, dx = -\operatorname{cosec} x + C$$

$$7. \int \tanh x \, dx = \ln |\cos hx| + C$$

$$\int \coth x \, dx = \ln |\sin hx| + C$$

$$9. \int \operatorname{sech} x \, dx = 2 \tan^{-1} \left(\tanh \frac{x}{2} \right) + C$$

$$10. \int \operatorname{cosech} x \, dx = \ln \left| \tanh \frac{x}{2} \right| + C$$

Practice Questions:

$$1. \text{ Prove that: a) } \int \sec x \, dx = \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C \quad \text{b) } \int \operatorname{cosec} x \, dx = \ln \left| \tan \left(\frac{x}{2} \right) \right| + C$$

$$2. \text{ Evaluate: a) } \int \frac{dx}{a+b \cos x} \quad \text{b) } \int \frac{dx}{a \sin x + b \cos x}$$

3. Evaluate:

$$\text{a) } \int \frac{dx}{\cos^2 x + 2 \sin^2 x} \quad \text{b) } \int \frac{dx}{a^2 \sin^2 x - b^2 \cos^2 x} \quad \text{c) } \int \frac{dx}{1 + 2 \sin^2 x} \quad \text{d) } \int \frac{dx}{(2 \sin x + 3 \cos x)^2}$$

4. Evaluate:

$$\text{a) } \int \frac{\cos x - \sin x}{\sqrt{\sin 2x}} \, dx \quad \text{b) } \int \frac{\sin x \cos x}{(\sin x + \cos x)^2} \, dx$$

5. Evaluate:

$$\text{a) } \int \frac{dx}{3+5 \cos x} \quad \text{b) } \int \frac{dx}{3 \sin x - 4 \cos x} \, dx \quad \text{c) } \int \frac{dx}{1 + \sin x + \cos x} \, dx$$

6. Evaluate:

$$\text{a) } \int \frac{dx}{\sin x + \cos x} \, dx \quad \text{b) } \int \frac{dx}{\cos x - \sqrt{3} \sin x} \, dx$$

7. Evaluate:

$$\text{a) } \int \frac{dx}{3+4 \cosh x} \quad \text{b) } \int \frac{dx}{1+3 \cosh^2 x} \, dx \quad \text{c) } \int \frac{\sinh x}{4 \tanh x - \operatorname{cosech} x \cdot \operatorname{sech} x} \, dx$$

Answers:

$$3a. \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) + C ; 3b) \frac{1}{2ab} \ln \left| \frac{a \tan x - b}{a \tan x + b} \right| + C$$

$$; 3c) \frac{1}{\sqrt{3}} \tan^{-1}(\sqrt{3} \tan x) + C ; 3d) -\frac{1}{2(2 \tan x + 3)} + C$$

$$4a) \ln |\cos x + \sin x + \sqrt{\sin 2x}| + C ; 4b) \frac{x}{2} +$$

$$\frac{1}{2(1+\tan x)} + c ; 5a) \frac{1}{4} \ln \left| \frac{2 + \tan \frac{x}{2}}{2 - \tan \frac{x}{2}} \right| + c ; 5b)$$

$$\frac{1}{5} \ln \left| \frac{2 \tan \frac{x}{2} - 1}{2 \tan \frac{x}{2} + 4} \right| + c$$

$$5c) \ln \left| \tan \left(\frac{x}{2} \right) \right| + C ; 6a) \frac{1}{\sqrt{2}} \ln \left| \tan \left(\frac{\pi}{8} + \frac{x}{2} \right) \right| + C$$

$$; 6b) -\frac{1}{2} \ln \left| \tan \left(\frac{\pi}{12} - \frac{x}{2} \right) \right| + C$$

$$6. \quad 7a) \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{\tanh \frac{x}{2}}{\sqrt{7}} \right) + c ; 7b) \frac{1}{4} \ln \left| \frac{2 + \tanh x}{2 - \tanh x} \right| + C ;$$

$$8. \quad 7c) \frac{1}{4} \sin hx + \frac{1}{16} \ln \left| \frac{2 \sinh x - 1}{2 \sinh x + 1} \right| + C$$

Thank You

Integrals of Rational Functions:

Rational Functions:

Consider two polynomial functions $f(x)$ and $g(x)$ such that $g(x) \neq 0$, then the function $\frac{f(x)}{g(x)}$ is called a rational objective function or rational fraction.

1. Proper Rational Fraction:

If degree of $f(x)$ is less than the degree of $g(x)$, then $\frac{f(x)}{g(x)}$ is called a proper rational fraction. Eg: $\frac{x^2+1}{x^3-1}$

2. Improper Rational Fraction:

If degree of $f(x)$ is greater than or equal to the degree of $g(x)$, then $\frac{f(x)}{g(x)}$ is called a proper rational fraction.

Example: $\frac{x^2+1}{x^2+x}$

Partial fractions:

Any proper rational function $\frac{f(x)}{g(x)}$ can be expressed as the sum of rational functions of simplest factor of $g(x)$, such fraction is called partial fractions.

Way of resolving into partial fractions

S.N	Factors in Denominator	Partial Fractions
1.	$(ax + b)(cx + d)$	$\frac{A}{(ax+b)} + \frac{B}{(cx+d)}$
2.	$(ax + b)(cx + d)(px + q)$	$\frac{A}{(ax+b)} + \frac{B}{(cx+d)} + \frac{C}{px+q}$
3.	$(ax + b)^2$	$\frac{A}{(ax+b)} + \frac{B}{(ax+b)^2}$
4.	$(ax + b)^3$	$\frac{A}{(ax+b)} + \frac{B}{(ax+b)^2} + \frac{C}{(ax+b)^3}$
5.	$(ax + b)(cx + d)^2$	$\frac{A}{(ax+b)} + \frac{B}{(cx+d)} + \frac{C}{(cx+d)^2}$
6.	$(px + q)(ax^2 + bx + c)$ Where $ax^2 + bx + c$ cannot be resolved into two linear factors)	$\frac{A}{(px+q)} + \frac{Bx+C}{(ax^2+bx+c)}$

Note:

In case of improper rational fraction such as $\frac{x^2+1}{x^2+x}$, firstly we resolve the numerator of improper rational function into proper rational fraction and then resolve into partial fraction:

Practice Questions:

1. Evaluate:

$$\begin{array}{llll} \text{a) } \int \frac{x}{(x-a)(x-b)} dx & \text{b) } \int \frac{2x+3}{x^2-2x-3} dx & \text{c) } \int \frac{x}{x^3+1} dx & \text{d) } \int \frac{x^2}{(x+2)(x+3)^2} dx \\ \text{e) } \int \frac{3x}{(x-1)(x^2+3)} dx & \text{f) } \int \frac{5}{(x+5)(2x^2+5)} dx & \text{g) } \int \frac{7x-10}{(3x-4)(x^2-2x+1)} dx & \text{h) } \int \frac{\sin x}{(\cos x+2)(\cos x+5)} dx \end{array}$$

Answers:

$$\begin{array}{llll} \text{(a) } \frac{1}{a-b} [a \ln |x-a| - b \ln |x-b|] + C & \text{(b) } -\frac{1}{4} \ln |x+1| + \frac{9}{4} \ln |x-3| + C & \text{(c) } -\frac{1}{3} \ln |x+1| + \frac{1}{6} \ln |x^2-x+1| + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) + C & \text{(d) } 4 \ln |x+2| - 3 \ln |x+3| + \frac{9}{x+3} + C \\ \text{(e) } \frac{1}{4} \ln |x-1| - \frac{1}{8} \ln |x^2+3| - \frac{1}{4\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) & \text{(f) } \frac{1}{11} \ln |x+5| - \frac{1}{22} \ln |2x^2+5| + \frac{\sqrt{10}}{11} \tan^{-1} \left(\sqrt{\frac{2}{5}} x \right) + C & & \\ \text{(g) } 2 \ln |x-1| - \frac{3}{x-1} - 2 \ln |3x-4| + C & \text{(h) } \frac{1}{3} \left| \frac{\cos x+5}{\cos x+2} \right| + C & & \end{array}$$

Some Special Types of Integrals

Type 1:

Integrals having only even powers of x

Rule: put $x^2 = y$ and resolve the rational fraction in y into partial fractions

Evaluate:

a) $\int \frac{dx}{(x^2+2)(x^2+4)}$ Ans: $\frac{1}{2\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} - \frac{1}{4} \tan^{-1} \frac{x}{2} + C$ b) $\int \frac{x^2 dx}{x^4-1}$ Ans: $\frac{1}{4} \ln \left| \frac{x-1}{x+1} \right| + \frac{1}{2} \tan^{-1} x + C$
 c) $\int \frac{x^2 dx}{x^4+2x^2-24}$ Ans: $\frac{\sqrt{6}}{10} \tan^{-1} \frac{x}{\sqrt{6}} + \frac{1}{10} \ln \left| \frac{x-2}{x+2} \right| + C$ d) $\int \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)}$ Ans: $\frac{1}{a^2-b^2} \left[a \tan^{-1} \frac{x}{a} - b \tan^{-1} \frac{x}{b} \right] + C$

Type 2:

Integrals having odd powers of x in numerator and even power in denominator

Rule: put $x^2 = y$ and differentiate w. r. to x. Resolve the integrand in y into partial fractions

Evaluate:

a) $\int \frac{x^3 dx}{(x^2+a^2)(x^2+b^2)}$ Ans: $\frac{1}{2(a^2-b^2)} [a^2 \ln |x^2 + a^2| - b^2 \ln |x^2 + b^2|] + C$ b) $\int \frac{x^3}{2x^4-3x^2-5} dx$
 Ans: $\frac{1}{14} \ln |x^2 + 1| + \frac{5}{28} \ln |2x^2 - 5| + C$ c) $\int \frac{2x^3+8x}{2x^4-5x^2+3} dx$ Ans: $\frac{11}{2} \ln |2x^2 - 3| - 5 \ln |x^2 - 1| + C$

Type 3:

Integrals of the form $\int \frac{x^2 \pm 1}{x^4+kx^2+1} dx$, where k is any integer (positive, negative or zero)

Rule: Divide N^r and D^r by x^2 and put $\left(x - \frac{1}{x}\right)$ or $\left(x + \frac{1}{x}\right) = y$,

which after differentiation gives N^r of the integrand.

Evaluate:

a) $\int \frac{x^2-1}{x^4+x^2+1} dx$ Ans: $\frac{1}{2} \ln \left| \frac{x^2-x+1}{x^2+x+1} \right| + C$ b) $\int \frac{x^2+1}{x^4-x^2+1} dx$ Ans: $\tan^{-1} \frac{x^2-1}{x} + C$

Type-4:

Integrals of the form $\int \frac{dx}{(x-a)^m(x-b)^n}$ where m and n are positive integers and $|a| < |b|$

Rule: Put $y = \frac{x-a}{x-b}$ and differentiate and change the integrand in terms of y

Evaluate:

a) $\int \frac{dx}{(x-2)^2(x-3)^3}$ Ans: $-\frac{1}{2} \left(\frac{x-2}{x-3}\right)^2 + 3 \left(\frac{x-2}{x-3}\right) - 3 \ln \left| \frac{x-2}{x-3} \right| - \frac{x-3}{x-2} + C$
 b) $\int \frac{dx}{(x+1)^2(x-2)^3}$ Ans: $-\frac{1}{81} \left[\frac{1}{2} \left(\frac{x+1}{x-2}\right)^2 + 3 \left(\frac{x+1}{x-2}\right) - 3 \ln \left| \frac{x+1}{x-2} \right| + \frac{x-2}{x+1} \right] + C$

Evaluate:

a) $\int \frac{x^3-1}{x^3+x} dx$ Ans: a) $x - \ln|x| + \frac{1}{2} \ln(x^2 + 1) - \tan^{-1}(x) + C$
 b) $\int \frac{x^4}{x^4-1} dx$ Ans: a) $x + \frac{1}{4} \ln \left| \frac{x-1}{x+1} \right| - \frac{1}{2} \tan^{-1}(x) + C$

Some Past Questions:

a) $\int \frac{8}{(2x-1)(2x+1)} dx$ b) $\int \frac{2x^2+3}{x^3+3x^2+2x} dx$ c) $\int \frac{1}{x^4+x^2+1} dx$ d) $\int \frac{x^2+8x+4}{x^3-4x} dx$ e) $\int \frac{1}{1+x+x^2+x^3} dx$
 Ans: a) $2 \ln \left| \frac{2x-1}{2x+1} \right| + C$ b) $\frac{3}{2} \ln x + \frac{11}{2} \ln|x+2| - 5 \ln|x+1| + C$ c) $\frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{x^2-1}{\sqrt{3}x} \right) - \frac{1}{4} \left(\frac{x^2-x+1}{x^2+x+1} \right)$
 d) $-\ln|x| + 3 \ln|x-2| - \ln|x+2| + c$ e) $\frac{1}{2} \ln|1+x| - \frac{1}{4} \ln|1+x^2| + \frac{1}{2} \tan^{-1} x + c$

Some Conceptual Questions:

1. Define proper and improper rational fractions with example of each.
2. Define partial fraction with example.
3. The integration of $\frac{x^3+x}{x-1}$ with respect to x requires concept of rational fractions by partial fraction. Justify it.